

Challenge (Habanero)

Q1. Evaluate the expression: $3 \cdot 2 + 11 - 5$

- (A) 15
- (B) 16
- (C) 17
- (D) 19
- (E) 21

Q2. A hiker's backpack weighs 8 pounds. If the hiker adds $2 \cdot 3$ energy bars, each weighing 1 pound, what is the new weight of the backpack?

- (A) 10
- (B) 11
- (C) 12
- (D) 13
- (E) 14

Q3. Translate the phrase '5 more than x ' into an algebraic expression

- (A) $x - 5$
- (B) $x + 5$
- (C) $5x$
- (D) $5 - x$
- (E) $x^2 + 5$

Q4. Simplify the expression: $2x + 3x - x$

- (A) $3x$
- (B) $4x$
- (C) $5x$
- (D) $6x$
- (E) x^2

Q5. A car's fuel consumption is given by $2 \cdot x + 15$, where x is

the number of miles driven. If the car has driven 7 miles, how much fuel has it consumed?

- (A) 20
- (B) 21
- (C) 29
- (D) 31
- (E) 33

Q6. Evaluate the expression: $(3 + 2)^2 - 1$

- (A) 15
- (B) 16
- (C) 24
- (D) 25
- (E) 30

Q7. The population of a species of rabbit is given by $5 \cdot x + 20$, where x is the number of months since the initial population was recorded. If $x = 4$, what is the current population of rabbits?

- (A) 40
- (B) 45
- (C) 50
- (D) 55
- (E) 60

Q8. Simplify the expression: $9 - 3 + 2 \cdot 4$

- (A) 10
- (B) 12
- (C) 14
- (D) 16
- (E) 18

Open Ended Questions (FRQ)

Q9. A park ranger observes the population of a species of deer and records the data as follows: the population is 25 deer initially and increases at a rate of $3 \cdot x$ deer per month, where x is the number of months since the initial observation. Write an expression for the population of deer after x months and evaluate it for $x = 6$. Provide your work and final answer.

Q10. A hiker is on a trail that has an elevation gain of 200 feet per mile. If the hiker has already hiked $2 \cdot x$ miles and wants to reach the summit, which is 1200 feet above the starting point, what is the value of x ? Write an equation representing the situation and solve for x . Provide your work and final answer.

Chef's Tips

- Emphasize the importance of following the order of operations when evaluating expressions
- Encourage students to use real-world scenarios to make algebra more relatable and interesting
- Remind students to simplify expressions by combining like terms